HGMC汇冠

200L

Brewpub Project Solution

To: Mr Filippo Greco
From: Mia Mu
Date: 9th Oct, 2026

Shandong HG Engineering Equipment Co., Ltd.



HGMC汇冠

S SPACESHIP

It is a high-end product developed for craft breweries. It has powerful functions, rich configurations, energy saving and consumption reduction, and good appearance.

C CASTLE

It is a standard product developed for craft breweries. It has excellent performance, flexible operation, durability and reliability, and high cost performance.

M MASTERBREW

It is a high-end product developed for the micro-craft brewing market. It adopts a modular design, suitable for multiple recipes, and a flexible pilot equipment that conforms to operating habits.

H H-STANDARD

It is a standard product developed for the micro-craft brewing market. It is simple, durable and cost-effective.

HIGH TECHNOLOGY

HIGH INTELLIGENCE

HIGH EFFICIENCY

HIGH COST PERFORMANCE

Equipment List and Quotation

NO.	Equipment	Qty	Specifications	Price (USD)
			Malt Mill System	
01	Malt mill	1 set	80-150kg/h, Double roller	360
			Brewhouse System	
01	200L Mash/Lauter tun	1 set	Electric heating	7025
02	200L Kettle/ Whirlpool tun	1 set	VFD raker,Milled sieve,Tangential inlet	
03	Non-Slip Work Platform	1 set	SS,Matched with brewhouse, non-slide	
04	Sanitary Centrifugal Pump	1 set	1m³/h, lift 18m	
05	Plate heat ex-changer	1 set	3 ㎡, Heating change area, double stage	
06	Indoor exhaust system	1 set	DN125	
07	Pipeline and accessories	1 pc	Full SS304 Pipe/Solenoid Valves/Wort Aeration Device/Measuring bucket/saccharimeter/grain remover/Wort Grant Column/ Feed water flow meter	
			Fermentation System	
01	200L Fermenters	3 sets	double layers with cooling jackets	5315
02	Yeast Adding Tank	1 set	2L ,SS304	
03	Beer hose	2 set	5 meters each	
04	Cooling and Insulation pipeline	1 set	Matched with fermenters, PPR	

Equipment List and Quotation

NO.	Equipment	Qty	Specifications	Price (USD)
		Cooling System		
01	Glycol liquor tank	1 set	300L,SS304	3221
02	Chiller	1 set	3HP, air cooling	
03	Glycol water Pump	1 set	Q=2m³/h H=28m	
04	Cooling and Insulation pipeline	-	Matched with glycol tank,PPR	
		CIP System		
01	Alkali tank	1 set	50L, Stainless steel, electric heating	1885
02	Disinfectant tank	1 set	50L, Stainless steel	
03	Washing pump	1 set	3=5m³/h H=24m	
04	Pipelines and Valves, and control panel	-	Stainless steel 304, Solenoid valves	
		Control System		
01	PLC Control Panel	1 set	Stainless steel, with necessary Cable tray	3200
Total Price (EXW Price)			**21,006USD**	

Total amount for the 200L brewery equipments: 21,006USD.

Notes:

1.This price is based on EXW Factory Price, 3Phase 380V 50Hz.

2.Payment: 50%TT before production, 50%TT before shipment.

3.This quotation is valid within 15 days.

4.Time of delivery of the brewery equipment: 60 working days.

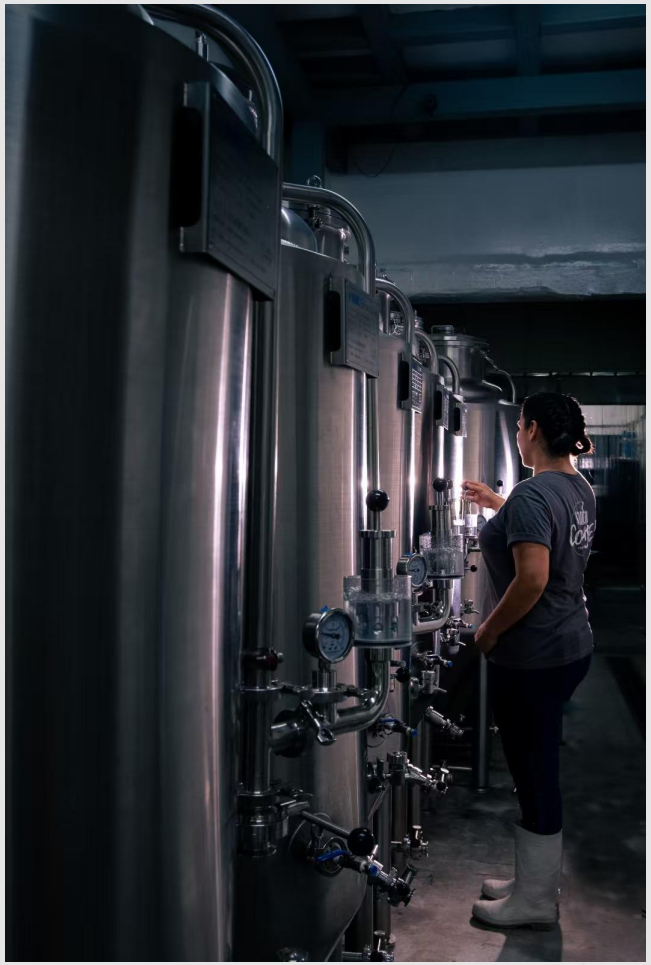
5. Guarantee: 3 years for the tanks and 1 year for the electrical components.

6. Layout advice, visually 3D drawing, individually flow chart design etc. are provided for free.

200L Brewery







Raw material processing system

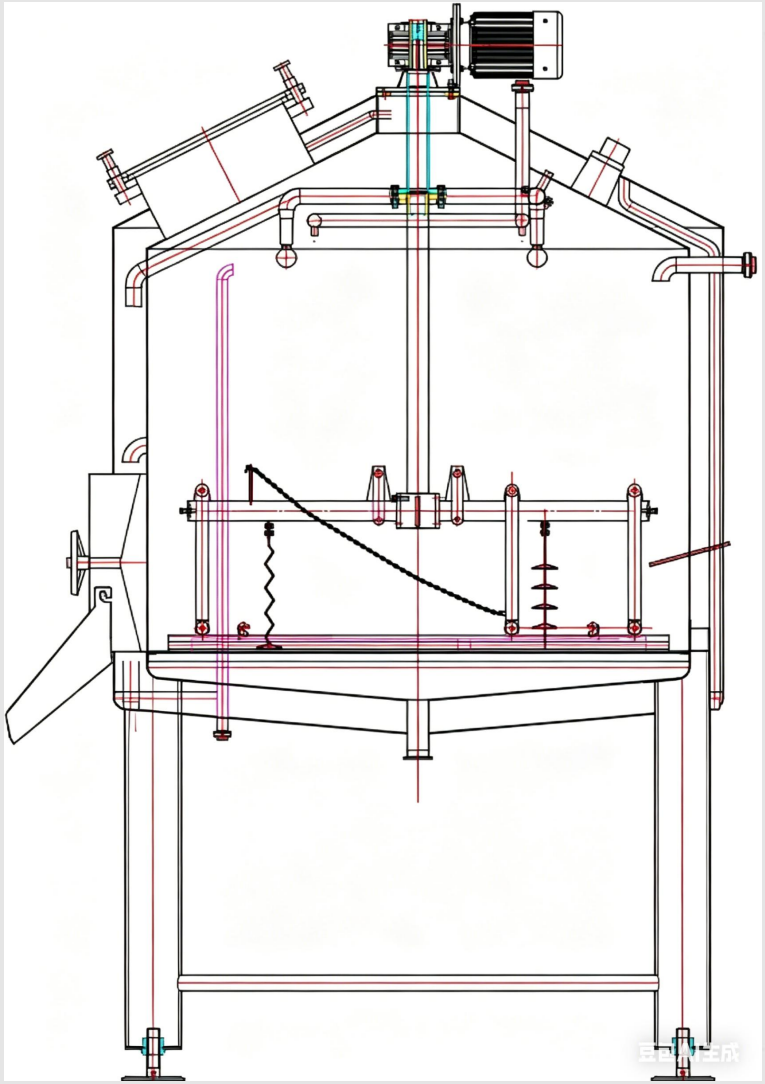
Malt mill



Production capacity: 80-150kg/h;
Equipment structure: roller type, motor, pulley, belt matching;
Motor power 1.5kw;
Material: stainless steel;
Roller size: 155 * 200mm
Roller speed: 700R/min asynchronous, adjustable gap
Temperature of crushed malt powder: less than 56 ℃
Dimensions: 700 * 680 * 1350 around
Equipment weight: 150Kg around
Usage: Grinding of malt, the main raw material for beer production

Mash system

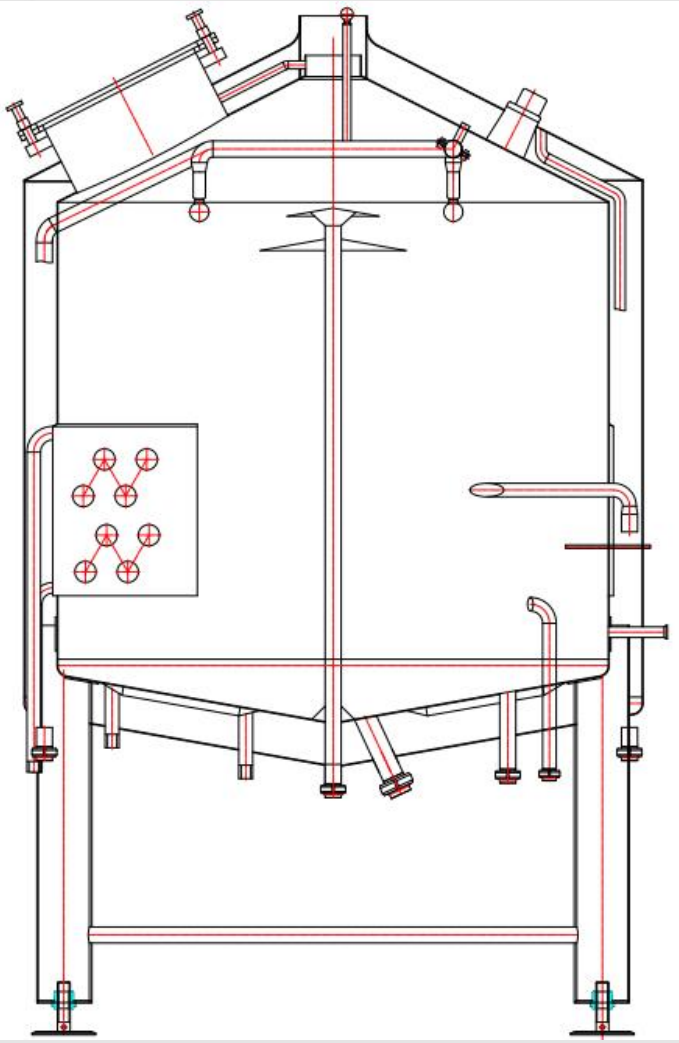
Mash/ Lauter Tun



Effective Volume: 200L; Total Volume: 320L; Vessel Diameter: 860mm; Total Height: 2200mm.
Filtration Area: 0.4 m². Filtration Speed: Wort filtration time: less than 100 minutes; Filtration rate: 6 L/min.
Agitator Testing: Installation complete; mechanical components subjected to a 2-hour test run.
Inner Shell Thickness: 3mm; Bottom Plate Thickness: 4mm; Outer Shell Thickness: 2mm; Material: 304 Stainless Steel.
Upper Head: Dish-shaped; Lower Cone Angle: 160°.
Internal Surface Finish: Inner shell pickled and passivated (Ra ≤ 0.8 μm); Outer surface brushed (satin finish); Welds polished with a 40mm sanding belt.
Cleaning Nozzles: Pin-connected spray balls; includes two rotating spray balls (with DN32 connections) for vessel cleaning.
Sight Glass/Manhole: Full-glass manhole (DN350); Integrated rechargeable LED light (DN80).
Motor Conduit: Electrical conduit (DN25) pre-embedded within the insulation layer.
Insulation Material: 80mm rock wool, clad in stainless steel.
False Bottom (Filtration Plate): Filtration slot width: 0.7mm; Open area ratio: 12%; Milled-slot sieve plate (4mm thick) designed to ensure thorough sparging of spent grains; Sieve plate flow area increased by over 20%. Sieve plate load capacity: 240 kg/m²; Clearance between sieve plate and the top of the lower cone: 30mm; Surface finish: Bright and flat; Features removable sections for easy cleaning.
Sparging Piping: Ring manifold located at the top of the filtration vessel; Features stainless steel spiral nozzles to ensure uniform distribution of sparge water and high spray density.
Motor & Agitator: Top-mounted 0.75 kW worm gear reducer; Speed ratio: 60:1; Model: NMRV063 (high service factor).
Rake/Plow Assembly: Angled rake blades designed to reduce resistance during grain bed agitation and facilitate efficient grain bed transport; Blade length: 350mm.
Vertically staggered rake blades ensure a uniform sparging process, preventing the formation of "channels" (preferential flow paths) that would allow sparge water to bypass the grain bed. Ensures thorough contact between the spent grain bed and sparge water, resulting in high yield and reduced filtration time.
Equipped with a manually operated spent grain scraper; side discharge of spent grain.
Spent Grain Discharge Mechanism: Side-mounted square spent grain manhole (430 × 330 mm).
Mash Piping: Side-mounted mash inlet with a pipe diameter of 2S (ϕ51 mm); material flow velocity ≤ 1.5 m/s.
Wort Outlet: Pipe diameter of 1.5S (ϕ38 mm); material flow velocity ≤ 1.5 m/s.
Automation: Temperature sensor included.
Support Structure: Leg supports (ϕ102 × 3 mm), featuring M16 × 80 mm adjustable leveling bolts and 38 mm lateral cross-bracing.
Key Features: High-throughput screen plates, cross-distributed raking blades, and a high-service-factor worm gear reducer.
Application: Facilitates the spent grain/liquid separation process.

Mash system

Kettl/ Whirlpool tun



Effective Volume: 200L; Total Volume: 320L; Vessel Diameter: 860mm; Total Height: 2200mm.
Heating Area: 0.7 m^2; Maximum Boiling Intensity: up to 8%; Heat Transfer Efficiency: ≥ 53 kW.
Whirlpool Inlet Velocity: 3.5–4 m/s; Tangential Entry Angle: 5–10°.
Inlet Flow Rate: 3 m^3/h; Pipe Diameter: 32mm.
Inner Shell Thickness: 3mm; Bottom Thickness: 4mm; Outer Shell Thickness: 2mm; Material: 304 Stainless Steel.
Upper Cone Angle: 126°; Lower Cone Angle: 160°.
Internal Surface: Inner shell pickled and passivated with smooth radius transitions (Ra ≤ 0.8 μm); External surface features a brushed finish with 40mm polished bands at the weld seams.
Condensate Drain Line: DN20, equipped with a float-type steam trap.
Cleaning Nozzles: Pin-connected spray balls; includes two rotating spray balls (32mm connection) for vessel cleaning and one fixed spray ball (12.7mm connection) for cleaning the riser pipe.
Glass Manhole: Full-glass design (DN350); includes a 10W/24V high-brightness illumination lamp (DN80) with an IP54 protection rating, compliant with VDE standards.
Indoor Steam Vent: Vent stack (ϕ133mm) equipped with a water spray cooling system.
Decorative Stack: ϕ133mm, Height: 1000mm, topped with a 400mm decorative cap.
Vent Condensate Recovery: The drain pipe (25mm connection) is embedded within the insulation layer.
Insulation Material: 80mm rock wool, clad in stainless steel. Forced Circulation: Features a large heat exchange surface area and controllable steam pressure, ensuring a low thermal load on the wort to preserve foam-forming proteins. A wort deflector hood creates an expanded evaporation surface, thereby increasing the wort evaporation rate. Forced circulation driven by a boiling pump enhances the overall heat transfer coefficient; a total evaporation rate of 8% is sufficient to meet process requirements.
An external spray ball is included to facilitate the cleaning of the vapor hood (umbrella cap).
Vessel Design: Optimized geometric dimensions.
Hop Addition: Hops are added via a dedicated hop dosing vessel.
Wort Piping: Bottom inlet connection, 1.5" Tri-Clamp (ϕ38).
Automation: Equipped with liquid level switches and temperature sensors.
Support Structure: Supported by legs (102 mm x 3 mm), reinforced with lateral cross-bracing.
Equipment Function: Executes the wort boiling process, driving off undesirable volatile compounds, concentrating the wort to a target density, facilitating the addition of hops and other adjuncts, and separating hot break (thermal coagulates).

Mash system

Plate Heat Exchanger



- 3m² single-stage six-pass
- Q=0.5KL/h，ΔH=0.1MPa，（3m² single-stage six-pass）
- Suspended type, wort inlet and outlet from top
- Wort inlet: 98℃, wort outlet: 8℃
- Brew water inlet: 4℃, hot water outlet: 82℃
- Wort inlet: ϕ38.1 clamp, wort outlet: ϕ38.1 clamp
- Brew water inlet: G1 external thread, brew water outlet: ϕ38.1 clamp
- **Equipment use:**
- Cool wort to a suitable temperature and recover heat from wort to hot water

Mash system

Brewhouse pipeline



- 38/32mm
- Grain Inlet & Sparge Water Pipeline (with check valve, mixing valve, regulating valve, flowmeter, thermometer): ϕ38
- Mash Transfer Pipeline: ϕ38
- Filtration Pipeline (with regulating valve, flowmeter, sampling station, level control, sight glass): ϕ38/32
- Wort Boiling Pipeline: ϕ38/ϕ32
- Whirlpool Wort Pipeline (Dual high/low wort outlets for time saving): ϕ38/ϕ32
- Plate Heat Exchanger Backflush Pipeline: ϕ38
- Brewing Water / Coolant Temperature Control Pipeline: ϕ38 / G1
- CIP Pipeline: ϕ32
- Process control uses manual valves. The main sanitary valves, pipelines, and pipe fittings (feed water valves, mash pipeline valves, water involved in the saccharification process, and wort) are all ISO standards

Mash system

Temperature Measurement and Oxygenation Combination Unit / Titanium Rod



Specifications:
Oxygenation Titanium Rod, L=125mm, 0.45μm
Bimetal Thermometer for on-site temperature display, WNG/WNY (0 ~ 100°C).
Clamp Connection, ϕ38.

Mash system

Measuring Bucket / Saccharimeter / Grain-Out Rake





Specifications:
Grain-Out Rake: Length 1000mm
Wort Measuring Cylinder: ϕ32, Height 350mm
Saccharimeter: One each of 10° and 20°

Mash system

Indoor Exhaust Device



Specifications:
Main Body: ϕ102, with a reducing diameter at the bottom for collection, ϕ38.
Built-in Water Spray Nozzle: DN15, connecting tube ϕ19.

Sanitary Centrifugal Mash Pump



Rated flow rate: 1 m^3/h
Rated lift: 18 m
Rated power: 0.75 kW (variable frequency control)
Open impeller: 1420 RPM
Connection size: 51/51 mm
Application: Rapidly transfer mash to a filter tank

Mash system

Enclosed Operating Platform



Specifications:
Platform Deck: Checkered plate with a mirror finish, with kickplate.
Platform Legs: Brushed stainless steel square tubes.
Guardrail: 51/19 round tube guardrail, compliant with safety requirements of GB 4053.3.

Fermentation system

Fermentation tank



Effective volume: 200L, total volume: 295L, tank diameter: 760mm, total height: 1900mm
Heat exchange area: 0.9㎡, average cooling rate: ≥0.5℃/hr
Inner tank pressure test: 48-hour pressure test with full water
Laser jacket inspection: Design inspection and 48-hour pressure test of the jacket
End cap model: THA1100*3
Inner thickness: 3mm, cone thickness: 4mm, 304 stainless steel; outer thickness: 2mm, 304 stainless steel
Lower cone angle: 60°
Argon arc welding: Double-gun butt welding of the inner liner ensures full penetration; single-sided welding and double-sided forming of fittings
Internal surface: Pickled and passivated inner, rounded transition with no dead corners, Ra≤0.6μm
Cooling jacket: Laser welding of side walls, 0.8mm stainless steel, high strength
Cooling method: Dimple jacket cooling, full-volume cooling rate of 0.5℃/h, refrigerant controlled by a solenoid valve
Refrigerant inlet pipe: Thick-walled DN25 interface, refrigerant requirement 1m³/h -5℃ 25% ethylene glycol solution
Temperature control: Refrigerant controlled by an angle seat valve, single-stage temperature control
Top pressure manhole: DN400 or side elliptical manhole: 440*340
Cleaning Nozzle: In-situ cleaning, pin-connected rotating ball cleaner, pressure requirement 0.2MPa, used for fermenter cleaning, with CO2 purge hose
Thermometer Interface: High-precision temperature sensor with welded sanitary sleeve (ϕ10)
Insulation Material: 80mm polyurethane, density not less than 42.5Kg/m³, outer stainless steel sheath
Sampling Valve: Metal-sealed plug valve DN25
Mechanical Pressure Regulator PVRV: 0.2-2.2bar, shock-resistant pressure gauge on CIP arm 0-0.6MPa ϕ60
Material outlet: ϕ38; Universal beer outlet ϕ63.5/ϕ38; CIP connector: ϕ32
Hop addition port: 4S (ϕ102), enabling pressurized dry hop addition
Automation: Temperature sensor interface ϕ10
Support type: Stainless steel legs at the bottom
Applications: Used for beer fermentation, maturation, and storage

Cooling system

Glycol liquor tank





Function: Glycol water supply for FV, CLT or wort cooling

- Effective volume: 300L, total capacity: 380L
- Inner jacket leakage test: leakage test with full water for 48 hours
- Thickness of inner jacket: 3mm, thickness of tank bottom: 3mm, thickness of external jacket: 2mm, SS304
- Adopt argon arc welding: the inner jacket double gun butt welding to ensure weld-through, pipe fittings single-sided welding double-sided molding
- Insulation Material: Polyurethane 80 mm, 42.5 Kg/m³
- Cooling Method: Direct cooling by refrigeration unit.
- Design Pressure of Vessel: Atmospheric pressure.
- Design Temperature of Vessel: -8° C.
- Fabrication Process: Argon arc welding/laser welding; weld joint efficiency: 0.85.
- Internal Surface Protection: Weld seams polished with 40 mm sanding strip.
- External Surface Protection: Brushed stainless-steel finish; weld seams polished and sanded, surface roughness Ra ≤ 0.8 μm.

Cooling system

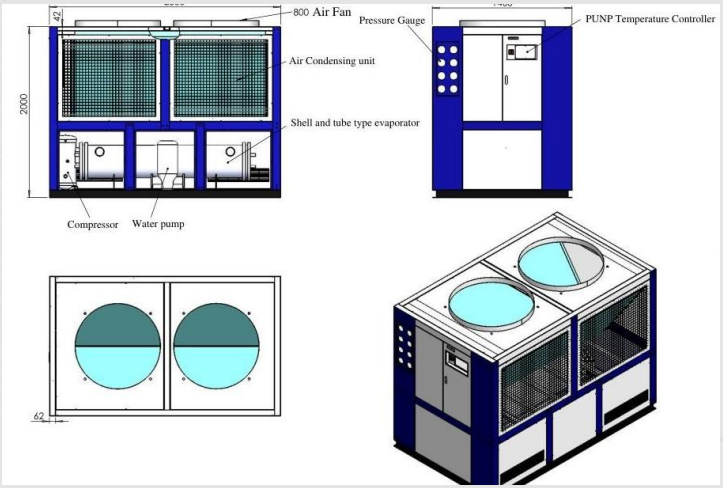
3HP Air-cooled chiller

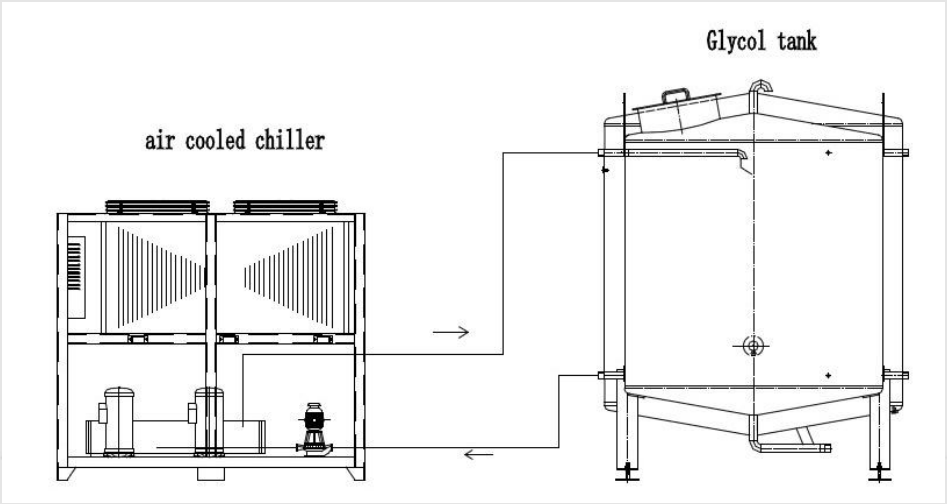




Technical parameters

- Air-Cooled Medium-Temperature Scroll Chiller (-5° C Outlet Water)
- Cooling Capacity: 5.7 kW / 3 HP
- Compressor Power: 2.81 kW
- Installed Power: 3.47 kW
- Refrigerant Piping: DN25, refrigerant temperature (0° C / -5° C)
- Dimensions: 1110*600*1200 mm
- Weight: 150 kg





Cooling system

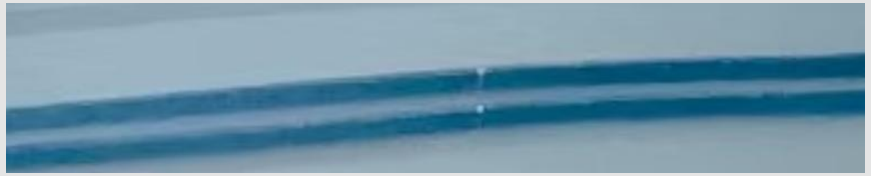
- 3m³/h Glycol liquor pump



Technical parameters

- Rated Flow: 2 m³/h
- Rated Head: 28 m
- Rated Power: 0.37 kW
- Open Impeller Speed: 1420 r/min
- Connection Size: G1 / G3/8
- Dimensions: 344*130*194 mm
- Application: Supply and circulation of refrigerant

- PPR pipeline



CIP system

50L Alkali tank



Effective volume: 50L, total volume: 65L, pot diameter 400mm, total height 1100mm
Container working medium: 2-3% alkaline solution
Inner liner thickness: 2mm, pot bottom thickness: 2mm, made of 304 steel plate material
Upper cone angle: 160 °, lower oblique base angle: 160 °
Processing technology: using argon arc welding, weld coefficient: 0.85
Internal Surface: Inner Tank Acid Pickled Passivation, Arc Transition, Ra≤0.8μm
The outer surface of the brushed weld seam is polished with 40mm sanding tape
Circular manhole: Top atmospheric manhole for easy maintenance. Exhaust: Integrated exhaust bend.
Temperature inside the tank: 40 ℃, controlled by a digital temperature gauge, electric heating power: 6kw
Hot pipeline: Manhole inlet and outlet pipes with a diameter of 1.5S (diameter 38)
Automation: Temperature sensors
Support form: Leg support, 76 * 2mm, with horizontal tension support

CIP system

50L Disinfectant tank



Effective volume: 50L, total volume: 65L, pot diameter 5=400mm, total height 1100mm
Inner liner thickness: 2mm, pot bottom thickness: 2mm, made of 304 steel plate material
Upper cone angle: 160 °, lower oblique base angle: 160 °
Processing technology: using argon arc welding, weld coefficient: 0.85
Internal Surface: Inner Tank Acid Pickled Passivation, Arc Transition, Ra≤0.8μm
The outer surface of the brushed weld seam is polished with 40mm sanding tape
Circular manhole: Top atmospheric manhole for easy maintenance Exhaust: Integrated exhaust bend
Manhole inlet and outlet pipes with a diameter of 1.5S (diameter 38)
Support form: Leg support, 76 * 2mm, with horizontal tension support

CIP system

CIP Pump & pipeline



Sanitary centrifugal pump
Rated flow rate: 5m ³/h
Rated head: 24 meters
Rated power: 1.5Kw variable frequency control
Open impeller: 1420R/min
Interface size: 38/38mm
Equipment purpose: Supply and circulation of CIP liquid
CIP pipeline: ϕ 38
The process control adopts butterfly valves, and the main sanitary valves, pipelines, and fittings are all ISO standards

Control System



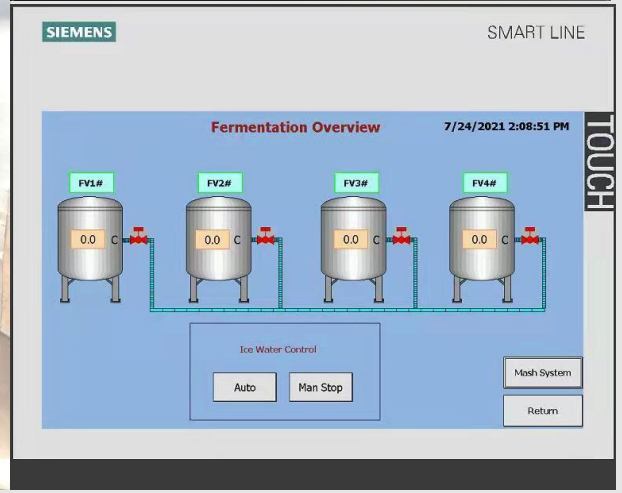
PLC Control System	
Automation	Fully automatic steam heating
	Fully automatic glycol water cooling
	Fully automatic pressure control
	Adjustable and controllable steam heating
CIP process	Fully automatic
Pumps & motors	VFD, intelligent mixing
Material automation & level switch	Add water to mash tun and mix with water (Option)
	Automatic temperature gradient recipe(Option)
	Adaptive frequency conversion motor(Option)
	Automatic pouring of mash(Option)
	Automatic plowing(Option)
	Automatic reflow(Option)

Control System









HGMC汇冠



• COMPANY STRENGTH

18+

18+ years industrial experience from malt milling to mashing and beer filling lines

TOP

The largest turnkey project supplier in North China, Top 3 in China.

28000+

28,000+ sqm plant and 300+ staff company scale

120+

120+ countries coverage, serving 5,000 customers globally

ISO...

ISO9001, CE, PED, UL, EAC, GOST certified

• TECHNOLOGY STRENGTH

- **3 members** having “Taishan Scholor” senior technical titles
- **5 members** with intermediate tiles.
- **6 members** with junior titles
- **20+** staff engaged in science and technical activities
- **79 patents**



HGMC



• The Netherlands-1000L



• China-500L



• Heineken-1000L



• Montenegro-2000L



• France-2000L



• Kona-1000L



• Australia-1500L



• Australia-2000L



• Sanya-2000L



• South Korea-3000L



• Canada-2000L



• Greece-1000L



• Ethiopia-1500L



• Russia-10Tons malting machine

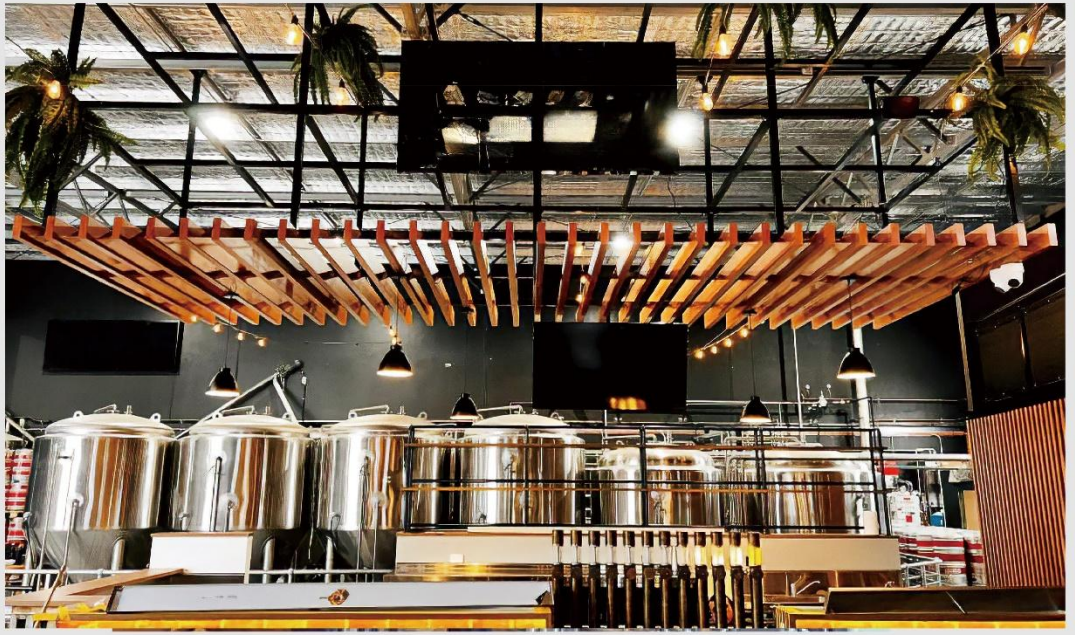


• Shenzhen-60KL

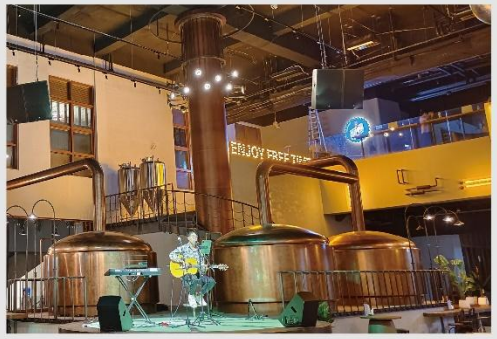












HGMC 汇冠

• AUTOMATIC LASER WELDING MACHINE

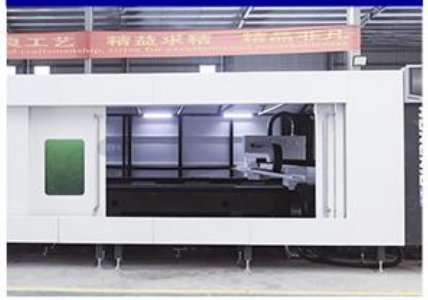


High pressure bearing capacity 1.5Mpa; The flow combination of glycol liquor is better matched; CNC automatic blanking and welding having higher precision; Automatic welding quality more stable.

FINISHED PRODUCT

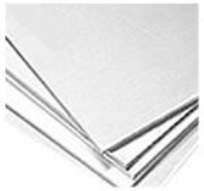


• LASER CUTTING MACHINE



Cut precisely as per technical drawing
Smooth cutting edges without burrs
3D cutting available to meet customized designs

FINISHED PRODUCT

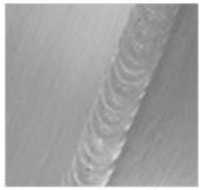


• AUTOMATIC WELDING MACHINE



Perfect and uniform longitudinal welding seams high tensile resistance, less oxidation and longer X-ray testing to avoid any welding defects. Fast cooling of welding seams to avoid deformations.

FINISHED PRODUCT



• AUTOMATIC PUNCHING MACHINE



Automatic formation of dimple jackets as to design uniform shaping and punching without damages.For high pressure testing and durability.

FINISHED PRODUCT



• CONE ROLLING MACHINE



Precise cone formation as per custom design
Protect the finish from scratching or damaging
To avoid any deformation failures

FINISHED PRODUCT



• POLISHING MACHINE



Well control of polishing precisions as requirements
High polishing accuracy to ensure sanitary finishes
Smooth and shiny appearance

FINISHED PRODUCT



• POLISHING DETECTOR



The hydrothermal tank reaches 0.6μm
The cold tank reaches 0.4μm

FINISHED PRODUCT



Look forward to cooperating with you!

From HGMC

Mia Mu